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SimCC coordinate based learning of human pose constraint information

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ABSTRACT

This paper proposes a novel approach to improve the accuracy of human pose estimation, which is a technique to predict the coordinates of human body keypoints in an input image. Existing methods often focus on keypoint localization without considering their positional relationships. We introduce two regularization terms based on SimCC coordinates: Bone loss and Sum loss of visible keypoints. They constrain the spatial relationship and visible number of keypoints respectively. To better leverage these loss functions, we enhance the deconvolution module with a combination of convolution and bilinear interpolation, which effectively recovers image resolution and preserves more feature map details. We conduct a comprehensive evaluation of our approach on the COCO dataset. Our approach enhances the accuracy of the top-down method by incorporating HRNet-W48. We achieve 77.7AP on the COCO validation set, which is competitive with state-of-the-art methods.

1. Introduction

Human pose estimation is an important task in computer vision, which is used to predict the coordinates of specific human body keypoints in the input image. It has a wide range of application areas such as human-computer interaction, security monitoring, animation special effects [29], 3D pose estimation [5], etc. 2D human pose estimation is the basis of these visual understanding tasks.

In recent years, many excellent pose estimation models have been proposed, including [25,32,38,40] of top-down method, [3,12,24,28,41] of bottom-up method, and [15,21] of end-to-end method. Not only the model structure, but also the pipeline of pose estimation has been explored in many aspects, such as data augmentation [8,9,11], coordinate coding [13,16,43], and loss function [20]. They have also effectively improved the accuracy of prediction without changing the model structure, but there are still many issues to be addressed.

Reference information from human pose estimation can be split into appearance information and constraint information. The keypoints are located using appearance information. The constraint information represents the relationship between the human body's keypoints or the interaction between the human body and the surrounding environment. It has important guiding significance when it is difficult to locate keypoints such as blurring or occlusion [9].

However, most of the existing studies focus on appearance information and ignore constraint information. As shown in Fig. 1, only part of the human body appears in the picture, but the network predicts redundant keypoints and unreasonable limb length. Which can result in poor judgment on downstream tasks such as action recognition. Thus, there is a need to increase the use of constraint information to help reduce erroneous estimations. In the human joint itself, there is much constraint information available, such as limited limb length and range of motion. However, it is more complex and diverse than appearance information, and it is also more difficult to learn the rules from the network. [3,39] enables the model to learn local information by designing a network structure at the feature level. This specific network structure, however, increases the complexity of inference. We use the loss function to optimize it, which is easy to be compatible with a variety of network architectures, and will not increase the storage and computing costs of extra memory in training and inferencing.

The loss function commonly used for human pose estimation evaluates the accuracy of localization at each keypoint, such as L1 Loss based on regression method and L2 Loss based on heatmap method. However, the internal structure of human pose has not been well evaluated [31,33]. In our analysis of human joint connection, we introduce bone length loss based on the SimCC coordinates, namely Bone Loss. It measures the bone length by computing the distance between connected keypoints in the image. It then compares the predicted and actual bone lengths to evaluate the keypoint localization accuracy from a relative distance perspective. This provides more useful feedback for model updating and learning. By learning both the absolute and rela-

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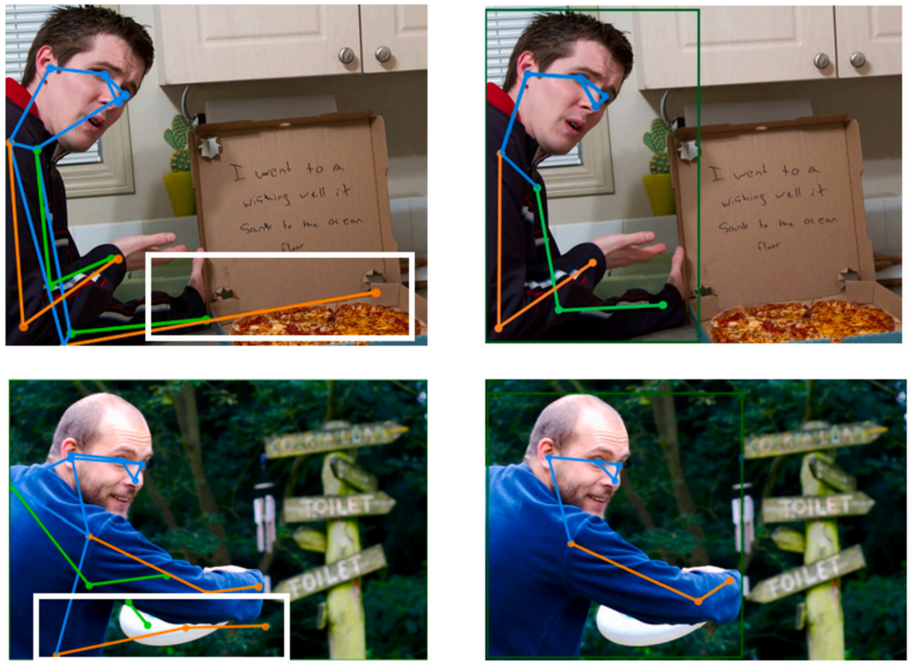


(a) redundant predictions (b) groundtruth

Fig. 1. The redundant keypoints of model predictions.

tive positions of keypoints, the model can avoid predicting unrealistic and non-existent poses.

We also study the problem of partial human body visibility in the image due to occlusion, varying shooting distances and data augmentation. This often leads to prediction errors in the number of visible keypoints, resulting in repeated or redundant keypoints that degrade the estimation quality. To tackle this issue, we introduce a simple constraint loss, called Sum Loss, that supervises the expected number of visible keypoints based on the image annotation information. The visibility of each keypoint is determined by its confidence level, which is easy to compute. By combining Bone Loss and Sum Loss with the position regression loss function, we obtain regularization terms that constrain the relative positions and number of visible keypoints. This enables the model to learn both the absolute and relative positions of keypoints and penalises inaccurate predictions more effectively without affecting well-recognised samples.

In addition, as part of the training process for the model, the feature map extracted from the backbone undergoes multiple upsampling layers to restore the low-to-high resolution before the loss function is computed. Restoring image resolution is the basis of calculating loss and coordinate decoding. This is a very important step but one that is often overlooked. Deconvolution (also known as Transposed convolution) is used in most models in the upsampling layer, but it suffers from some issues. For the deconvolution module, it is necessary to reasonably set super parameters such as kernel size and step size, and the feature map with improper parameter learning will make feature map have uneven overlap (checkerboard phenomenon), which is unavoidable for neural network [27]. A better method to ameliorate this problem is to separate convolution at higher resolution upsampling to compute the features. We have designed a module that first uses a convolution layer multiple times and then performs bilinear interpolation to adjust the size of the image, and obtained good results.

In this paper, we make the following contributions: (1) We introduce Bone loss based on SimCC coordinates as a regularization term for the loss function. This term helps to regulate the predicted bone length and prevent unrealistic limb length. (2) We propose Sum loss as another regularization term to limit the prediction of redundant keypoints. These two terms can be easily integrated with different algorithm models without increasing memory and computation costs. (3) We design an upsampling module that combines convolution and bilinear interpolation. This module preserves fine details in the feature map during the resolution restoration process.

Our method has been tested on multiple backbones, and ablation experiments have been performed on each proposed method using COCO and MPII datasets. The experiments show that the accuracy of prediction has been improved on multiple baseline models and different image input sizes.

2. Related work

2.1. Multi-person human pose estimation

There are three main methods for 2D multi-person human pose estimation: top-down methods, bottom-up methods, and single-stage methods.

Top-down methods involve using a human detector to identify all individual instances in the image. The instances are then cropped and resized to ensure they are of the same size before being input into the keypoint detector for individual pose estimation. The representative work of the top-down method includes Hourglass [25], SimpleBaseline [38], HRNet [32], ViTPose [40], etc. In general, the performance of the top-down method relies on the detector, and the inference speed is influenced by the number of instances present in the image. However, this method offers high accuracy in keypoint detection.

Bottom-up methods first detect all potential unidentified keypoints in the image, and then group them to associate the keypoints belonging to the same person. Classic methods include Openpose [3], Pifpaf [12], Associative embedding [24], and PersonLab [28]. The bottom-up method does not require manual detectors and can decouple the running time from the number of individuals, making it faster than the top-down method. However, its accuracy is generally lower due to the significant disparity between the grouping problem and the human scale.

Single-stage methods, such as FcPose [22], Directpose [35], PETR [30], and Poseur [21], have emerged in recent years. The single-stage method estimates the multi-person pose directly from the input image, simplifying the multi-person pose estimation pipeline. However, its accuracy is typically lower than that of top-down methods.

Our objective is to examine the constraint information between the joints of the human body. As the keypoint detection component of the top-down method only needs to predict the pose of a single individual, it is easier to identify and learn about information such as the connections and distances between the keypoints of a single human body. Therefore, this study is based on the top-down method.

2.2. Coordinate representation

The coordinate representation of keypoints can also be divided into three methods: Regression-based methods, Heatmap-based methods and SimCC-based methods.

Regression-based methods [4,35–37] directly supervise the model to learn coordinate values, i.e. the input image is mapped directly to body joint coordinates. This approach has a low computational cost and memory overhead and is end-to-end differentiable, making it friendly to devices with weak computing power. However, it is unable to learn spatial and contextual information and lacks robustness to complex situations, so performance is poor.

Heatmap-based methods: [3,17,32,38,40]. This approach represents each keypoint as a heatmap, where each element in the heatmap represents the confidence of the corresponding keypoint at that pixel in the image. Confidence can be seen as a way to describe the probability distribution of the model for the keypoint's location, the distribution is typically assumed to be a Gaussian or Laplace distribution. We usually select the pixel with the highest probability in the heatmap as the model's prediction. However, it's important to note that the operation of selecting the index with the maximum value, known as argmax, is non-differentiable, making it impossible to propagate gradients through it during model training. To address this issue, some methods like Integral Regression [34] link heatmap representation with regression. This approach modifies the "select maximum" operation to "take the expectation," estimating keypoints as integrals of all positions in the heatmap, weighted by their probabilities, a process referred to as softargmax. Methods such as DSNT [26] introduce variance regularization and distribution regularization in the loss function to impose prior constraints on the probability distribution for improved learning. Approaches like RLE [13] and Sampling-Argmax [14] also enhance learning of the probability distribution, leading to significantly improved accuracy in keypoint predictions.

SimCC [16] proposed a new coordinate coding scheme. It represents the coordinate estimation task as two classification tasks of horizontal and vertical coordinates. Specifically, it encodes the x and y coordinates of keypoints into two independent one-dimensional vectors with a higher quantization level to supervise learning. Through a scaling factor k (≥1), the length of the one-dimensional vector is greater than or equal to the edge length of the picture. This method achieves sub-pixel positioning accuracy, and alleviates quantization error, but does not need large memory overhead like heatmap. It provides a lighter and more effectively coordinate representation method for human pose estimation. Because of its convenience and accuracy, our proposed regularization term is computed based on SimCC coordinates.

2.3. Regularization

Regularization refers to modifying a learning algorithm so that it reduces the generalization error rather than the training error. Many different forms of regularization strategies can be used in deep learning work. Indeed, the development of more efficient regularization strategies has become one of the main research efforts in this area, and large numbers of tasks can likely be solved efficiently using some form of regularization [6]. A number of regularization strategies are available, with common ones being norm penalty, data augmentation, semi-supervised learning, and multi-task learning, etc. There are a number of strategies that add additional constraints to the deep learning model that constrain parameter values. In some strategies, additional terms are added to the objective function to place soft constraints on the parameter values. These additional constraints and penalties can improve model performance on the test set if we make a careful choice.

To regularize a learning model, we can add regularization terms to the loss function that reflect some prior knowledge or constraints. For example, [20] added a regularization term to constrain the heatmap's standard deviation of keypoints, while [31] added a regularization term to regularize the relative locations of each fashion landmark. In this paper, we propose two terms that penalize the deviation from the expected spatial relationship and number of visible keypoints in the human pose.

3. Methods

3.1. Overall framework

Fig. 2 shows the network framework diagram. The model first preprocesses the original image and then uses the backbone network to extract features. To reduce computational complexity, the feature map has a lower resolution than the original image by a certain factor. The model then restores the resolution of the feature map at the head and predicts the coordinate coding. The model learns from the loss between the predicted coordinates and the ground truth coordinates. Our research mainly focuses on improving the resolution recovery module and the loss function, which can effectively enhance the prediction accuracy of the model.

3.2. Resolution recovery module

The process of restoring the resolution of the feature map to the resolution of the original image is called upsampling. The upsampling in most studies uses the method of deconvolution. It is a classic deconvolution blocks, as shown in Fig. 2(a), consisting of a deconvolution layer followed by batch normalization [10] and ReLU [1]. A more efficient upsampling scheme was attempted. In particular, convolution with kernel sizes of 3×3 and 1×1 was carried out, followed by batch normalization and ReLU, as shown in Fig. 2(b). This block was stacked twice and the number of channels was adjusted in the second convolution layer of size 3×3. We then perform bilinear interpolation for upsampling by a factor of 2, and denote the output features of the base network as $F_{out} \in R^{\frac{H}{d} \times \frac{W}{d} \times C}$. H and W are the dimensions of the image input backbone, d is the downsampling multiplier, and C is the number of channels in the backbone output. The resolution restoration module i.e.

$$
\begin{aligned}
K' &= (ReLU(BatchNorm(Conv_{1\times1}(Conv_{3\times3}(F_{out}))),\\
K &= (Bilinear(K')).
\end{aligned}
\tag{1}
$$

$K \in R^{\frac{2nH}{d} \times \frac{2nW}{d} \times N}$ denotes the heatmap during upsampling, n represents the stacking times of the resolution recovery module and N denotes the number of channels. In the subsequent final layer, the number of channels is adjusted based on the number of keypoints to be estimated, which was set to 17 for the MS COCO dataset [19].

3.3. Bone length loss

Bone length loss (Bone loss) aims to constrain the relative positions between joints. Assuming that the human body defines N keypoints, the human pose estimation predicts the positions of all keypoints. The joint sitting predicted by Simcc is labeled $\hat{p} = \left(\hat{o}_x, \hat{o}_y\right)$,

$$
\hat{o}_x = \frac{argmax_i(o_x(i))}{k}, \hat{o}_y = \frac{argmax_j(o_y(j))}{k}, \tag{2}
$$

where $i \in \{0, 1, ..., W \cdot k - 1\}, j \in \{0, 1, ..., H \cdot k - 1\}$. k is the splitting factor of SimCC coordinates and is usually set to 2. Bone loss requires

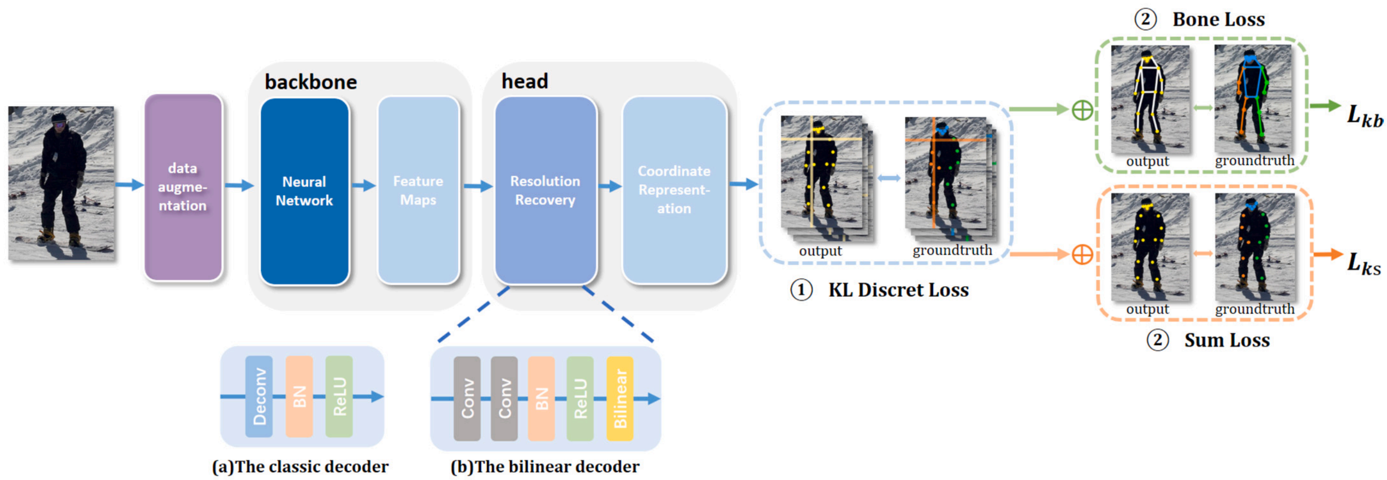


Fig. 2. The details of our framework.

calculating the distance between two coordinates. SimCC uses argmax to locate the coordinates of keypoints, but argmax is non-differentiable. Therefore, the loss function designed using argmax cannot be trained end-to-end in the model. To address this issue, we adopted softargmax to approximate the coordinates of keypoints computed using argmax. The calculation formula for softargmax is as follows:

$$softargmax(z) = \sum_i \frac{e^{\alpha z_i}}{\sum_j e^{\alpha z_j}} i, \tag{3}$$

for a vector z, where i represents the index of elements in z, and z_i represents the confidence score of the i^{th} element. Here, we use $\alpha = 10$, which increases the score of the maximum value and decreases the score of smaller values after weighted averaging, resulting in more accurate positional coordinates. This formula computes a weighted average of the element indices to obtain a continuous estimate of position. Consequently, softargmax is differentiable, making it suitable for neural networks and allowing for gradient computation and backpropagation. So, our formula for calculating the coordinates is:

$$\hat{p} = (\hat{o}_x, \hat{o}_y) = \left(\frac{softargmax_i(o_x(i))}{k}, \frac{softargmax_j(o_y(j))}{k} \right) \tag{4}$$

The coordinates predicted of the i^{th} keypoint are $\hat{p}_i$ and the confidence $\hat{v}_i$ is predicted for the visibility of each keypoint. We only perform the computation on visible keypoints. The confidence $\hat{v}_i$ is considered visible if the value of the keypoint exceeds a certain threshold, that is $\hat{v}_i = 1$, otherwise $\hat{v}_i = 0$. Therefore, combining the confidence levels of the keypoints, the predicted keypoint coordinates are

$$P_i = \hat{p}_i \cdot \hat{v}_i. \tag{5}$$

Bone length is the relationship between the positions of two joints. The distance between joints i and j can be defined as $\left\| P_i - P_j \right\|$, where i and j are the connected pairs of joints defined in the dataset. Bone loss is then the L1 loss between the predicted and true distances between two joints, with the true distance being calculated from the groundtruth labeled coordinates y_i and the visibility v_i. The true coordinate of a keypoint is

$$Y_i = y_i \cdot v_i. \tag{6}$$

So the formula for Bone loss is

$$L_{bone} = \sum_{i,j} \left| \left\| P_i - P_j \right\| - \left\| Y_i - Y_j \right\| \right|. \tag{7}$$

3.4. Sum loss

The number of visible keypoints for each person in 2D human pose estimation is an important piece of information. However, due to factors such as occlusion, distance, and pose variation, not all N body keypoints are visible in the image. Therefore, we need to impose more constraints on the number of keypoints and penalize incorrect predictions. We use both the model inference confidence and the dataset annotation information to compute this regularization term. The Sum loss equation we designed is as follows.

$$L_{sum} = \left| \sum_{i=1}^{N} \hat{v}_i - \sum_{i=1}^{N} v_i \right|, \tag{8}$$

which represents the L1 loss of the expected number of visible joints relative to the actual number of visible joints.

3.5. Overall loss function

SimCC makes use of the Kullback-Leibler divergence [23] as the loss function for model training, and both of our proposed methods can be combined with KL divergence separately. These two components of the loss function can be thought of as the regression loss and the regularization loss respectively, and for inaccurate predictions, more penalties will strengthen their fit position. The regularization loss using Bone loss is

$$L_{kb} = L_{kl} + \lambda L_{bone}, \tag{9}$$

where the KL divergence normalizes the absolute position of the keypoints and Bone loss normalizes the relative position. $\lambda \in [0, \infty]$ is a hyperparameter that weighs the relative contribution of the penalty term L_{bone} and the standard objective function L_{kl}. Setting λ to 0 indicates no regularization, with larger a corresponding to larger regularization penalties. Note that L_{bone} can be a large number but should not exceed L_{kl} value, the hyperparameter λ should therefore be set as small as possible in order to maintain equilibrium. Experimentally, we set it to 0.05. The loss function using Sum loss is:

$$L_{ks} = L_{kl} + \gamma L_{sum}. \tag{10}$$

$\gamma \in [0, \infty]$ is the weight of the regularization term L_{sum}. Through experiments, we set it to 0.5.

4. Experiment

In this section, we conducted an evaluation of our approach on two widely used datasets, COCO 2017 [19] and MPII [2]. Firstly, we will introduce the fundamental information and evaluation metrics of these two datasets, along with the specific details of our experiments. Subsequently, we will present and analyze the performance comparison between our method and baseline models across multiple datasets. Additionally, we conducted extensive ablation studies to validate the effectiveness of each proposed module. Finally, we visualize the comparison of pose estimation results between our proposed method and baseline models.

4.1. Experiment settings

4.1.1. Dataset

COCO[19] is a large-scale image dataset that contains various visual tasks, such as object detection, segmentation, keypoint detection, etc. Among them, the COCO keypoint detection dataset is a subset specifically used for human pose estimation, which contains more than 200,000 images and 250,000 human instances, each with 17 keypoints annotated. Our model utilizes the COCO 2017 training set, validation set, and test-dev set, which consist of 57,000, 5,000, and 20,000 images, respectively. The training set is employed for model learning and optimization, while the validation set and test-dev set are used for model evaluation and comparison.

MPII Human Pose Dataset [2] serves as a state-of-the-art benchmark for evaluating articulated human pose estimation. This dataset comprises approximately 25,000 images, encompassing over 40,000 individuals, each annotated with 16 key body joints. The images were systematically gathered, covering a wide range of everyday human activities. We trained our model on the MPII training set and evaluated its performance on the validation set.

4.1.2. Evaluation metrics

AP (Accuracy precision) serves as the evaluation metric for the COCO dataset. We consider a keypoint to be correctly localized if the similarity between the true keypoint and the predicted keypoint is greater than a threshold. AP represents the percentage of keypoints correctly located out of all the predicted keypoints. Object Keypoint Similarity (OKS) is used to evaluate the similarity of the keypoints between two individuals.

$$OKS = \frac{\sum_i exp(-d_i^2/2s^2k_i^2)\delta(v_i > 0)}{\sum_i \delta(v_i > 0)},$$

where d_i is the Euclidean distance from the detected keypoint to its corresponding groundtruth, s refers to the size of the figure, computed from the person box in the groundtruth, v_i is the groundtruth visibility flag, $v_i = 1$ means that this keypoint is visible in the image, and k_i refers to the normalization factor of the i^{th} skeleton point, reflecting the degree of influence of the current skeleton point relative to the whole.

PCKh (Percentage of Correct Keypoints with a Half-Body) is a metric used by the MPII dataset to assess the performance of human pose estimation. This metric requires calculating the distance between predicted keypoints and ground truth keypoints and comparing it to a predefined threshold. If the distance is less than the threshold, the keypoints are considered correctly estimated; otherwise, they are considered incorrect. Subsequently, the percentage of correctly estimated keypoints is calculated relative to the total number of keypoints. A smaller threshold demands more precise keypoint estimations, while a larger threshold allows for greater tolerance. Typically, PCKh@0.5 denotes a threshold set at half the body width, and PCKh@0.1 denotes a threshold set at one-tenth of the body width.

4.1.3. Baseline

We choose SimpleBaseline [38] with ResNet50 [7] as the backbone network, HRNet-W32 [32] and HRNet-W48 [32] models. The coordinates are represented using the SimCC [16] method. The experiments are conducted using image instances with input sizes of 64×64, 128×128, 256×192, and 384×288 pixels.

4.1.4. Implementation details

We trained our model using the Adam optimizer on a single GPU. For the baseline, we followed the original parameters outlined in the paper. Specifically, for the SimpleBaseline model [38], we used ResNet50 as the backbone, and the base learning rate was set to 1e-3. The learning rate was decreased to 1e-4 and 1e-5 at the 90th and 120th epochs, respectively, and we trained the model for a total of 140 epochs. For the HRNet-W32 and HRNet-W48 model [32], we set the base learning rate to 1e-3, which was decreased to 1e-4 and 1e-5 at the 170th and 200th epochs, respectively. We trained the model for a total of 210 epochs. To augment the data, we followed the settings of SimCC [16] and applied random flip, half body transform, and random scale rotation.

4.2. Analysis

4.2.1. Results on the COCO validation set

The results of our method and other state-of-the-art methods are presented in Table 1. We categorized the existing approaches into three types: regression-based, heatmap-based, and SimCC-based methods. Our method consistently outperforms other methods for different backbone networks at input sizes of 256×192 and 384×288. Using an image input size of 256×192, HRNet-W48 achieves an AP of 77.5 on the COCO validation set. By applying our proposed method, we boost its AP to 77.7, which outperforms many state-of-the-art methods. Similar improvements are also observed on SimpleBaseline and HRNet-W32 models.

4.2.2. Results on the COCO test-dev set

Table 2 presents a comparison of results on the COCO test-dev set using input sizes of 256×192 with HRNet-W32 and HRNet-W48 as backbone networks. Since the testing results of HRNet-w32 based on SimCC on the COCO test-dev dataset were not provided in the SimCC paper, the data for this algorithm in the table represents results obtained through training using their publicly available code. When our approach was applied to HRNet-w32 based on SimCC, it achieved an improvement of 0.3 AP. When applied to HRNet-W48, it reached an AP of 74.8.

4.2.3. Results on the MPII validation set

We conducted experiments on the MPII dataset, utilizing input sizes of 64×64 and 256×256, respectively. In Table 3, our approach led to an increase of 0.2 points in the average PCKh@0.5 score and an improvement of 0.4 points in the average PCKh@0.1 score. By examining the various metrics in Table 3, we observed that, as a result of the constraints imposed by our loss function, body joints prone to misidentification due to occlusion or truncation, such as the knee and ankle, exhibited significantly enhanced accuracy compared to other body parts. This also validates that our proposed approach has, to a certain extent, alleviated the issue we aimed to address.

4.2.4. Ablation study

We performed ablation experiments to evaluate the impact of the resolution restoration module, Bone loss, and Sum loss. We use SimpleBaseline as the base model and test it on COCO val 2017 dataset. The image input size is 256×192. Table 4 shows that using bilinear interpolation instead of deconvolution improves AP by +0.3. Using Bone loss or Sum loss alone increases AP by +0.1 each. Combining these two losses with bilinear interpolation results in an overall improvement of +0.6 AP, which indicates a synergistic effect.

4.2.5. Analysis of loss function

To verify the effectiveness of the two loss functions, we use Bone loss and Sum loss separately for different input sizes in the HRNet-W32

Table 1
Comparison with regression-based and heatmap-based state-of-the-art methods on the COCO val 2017 set.

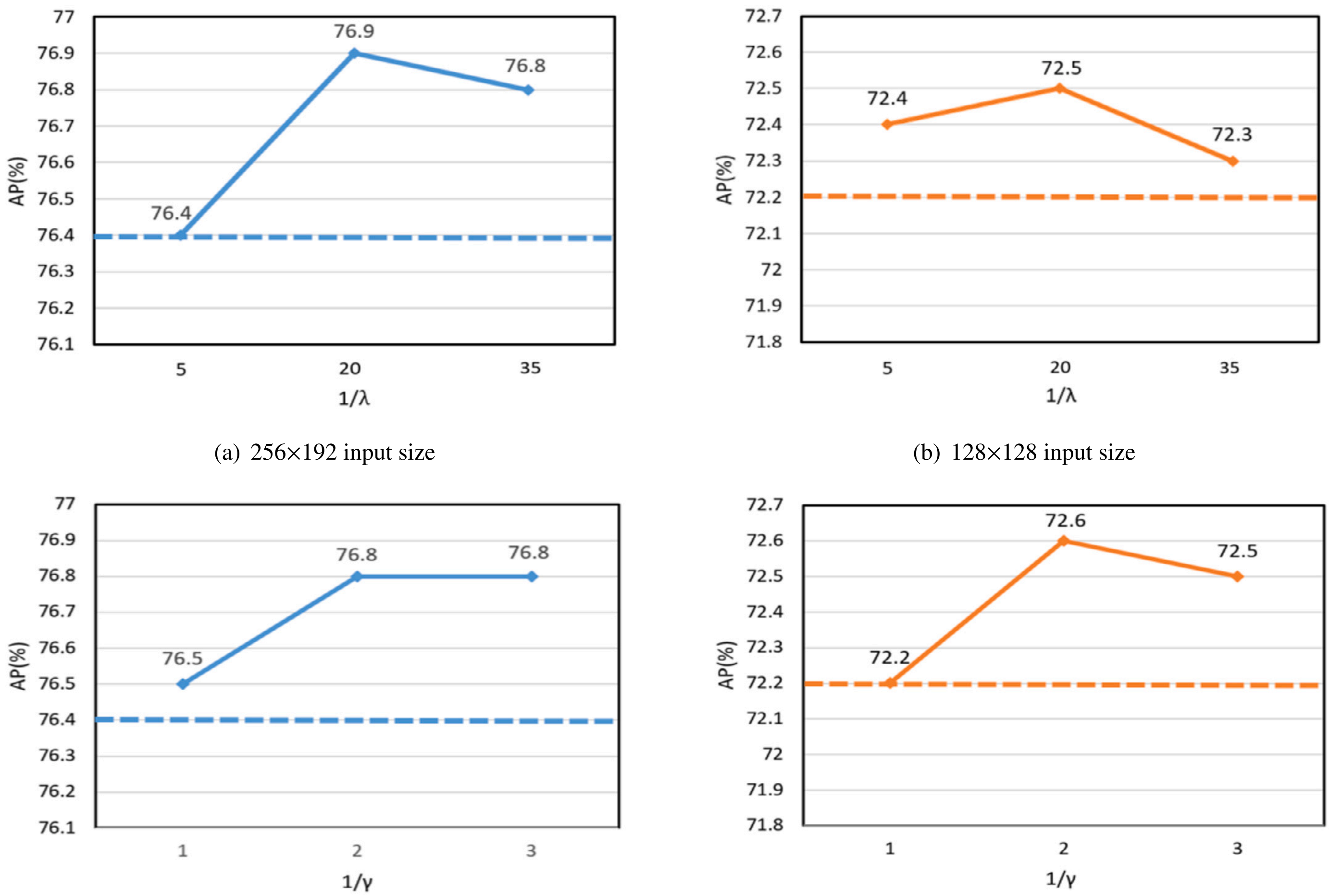
Method	Backbone	Input size	AP	AP^{50}	AP^{75}	AP^{M}	AP^{L}	AR
Regression-based								
PRTR [15]	HRNet-W32	384×288	73.1	89.4	79.8	68.8	80.4	-
PRTR [15]	HRNet-W32	512×384	73.3	89.2	79.9	69.0	80.9	-
RLE [13]	ResNet-50	256×192	70.5	88.5	77.4	-	-	-
RLE [13]	HRNet-W32	256×192	74.3	89.7	80.8	-	-	-
Poseur [21]	ResNet-50	256×192	75.4	90.5	82.2	68.1	78.6	-
Heatmap-based								
HRFormer-S [42]	HRFormer-S	256×192	74.0	90.2	81.2	70.4	80.7	79.4
HRNet-W32 [32]	HRNet-W32	256×192	74.4	90.5	81.9	70.8	81.0	80.0
ViTPose-B [40]	ViT-B	256×192	75.1	-	-	-	-	80.3
HRNet-W32 [32]	HRNet-W32	384×288	74.9	92.5	82.8	71.3	80.9	80.1
HRNet-W48 [32]	HRNet-W48	384×288	75.5	92.5	83.3	71.9	81.5	80.5
SimCC-based								
SimpleBaseline [38]	ResNet-50	256×192	73.8	92.4	80.3	71.0	78.6	77.0
+Ours	ResNet-50	256×192	**74.4**	**92.4**	**81.2**	**71.3**	**79.2**	**77.4**
SimpleBaseline [38]	ResNet-50	384×288	75.3	92.4	82.2	72.0	80.7	78.4
+Ours	ResNet-50	384×288	**75.6**	**92.4**	**82.3**	**72.2**	**80.8**	**78.6**
HRNet-W32 [32]	HRNet-W32	256×192	76.4	92.5	83.4	73.8	80.9	79.4
+Ours	HRNet-W32	256×192	**76.9**	**93.4**	**83.4**	**74.2**	**81.1**	**79.8**
HRNet-W48 [32]	HRNet-W48	256×192	77.5	93.4	84.4	74.7	82.4	80.5
+Ours	HRNet-W48	256×192	**77.7**	**93.4**	**84.5**	**74.8**	**82.5**	**80.5**

Table 2
Comparison with state-of-the-art methods on the COCO test-dev 2017 set.

Method	Backbone	Input size	AP	AP^{50}	AP^{75}	AP^{M}	AP^{L}	AR
Regression-based								
DirectPose [35]	ResNet-50	-	62.2	86.4	68.2	56.7	69.8	-
PointSetNet [37]	HRNet-W48	-	68.7	89.9	76.3	64.8	75.3	-
IntegralPose [34]	ResNet-101	256×256	67.8	88.2	74.8	63.9	74.0	-
PRTR [15]	HRNet-W32+Trans.	384×288	71.7	90.6	79.6	67.6	78.4	78.8
PRTR [15]	HRNet-W32+Trans.	512×384	72.1	90.4	79.6	68.1	79.0	79.4
Heatmap-based								
SimpleBaseline [38]	ResNet-50	384×288	71.5	91.1	78.7	67.8	78.0	76.9
SimpleBaseline [38]	ResNet-152	384×288	73.7	91.9	81.1	70.3	80.0	79.0
HRNet-W32 [32]	HRNet-W32	256×192	72.9	91.8	81.2	69.8	78.7	78.5
HRNet-W48 [32]	HRNet-W48	256×192	73.4	91.9	81.6	70.1	79.3	79.0
IA-H32 [39]	HRNet-W32	256×192	73.9	92.2	82.3	70.7	79.7	79.3
IA-H48 [39]	HRNet-W48	256×192	74.3	92.3	82.3	71.2	79.9	79.7
SimCC-based								
HRNet-W32 [32]	HRNet-W32	256×192	73.8	92.0	81.7	70.7	79.5	79.4
+Ours	HRNet-W32	256×192	**74.1**	**92.0**	**81.8**	**70.9**	**79.9**	**79.6**
+Ours	HRNet-W48	256×192	**74.8**	**92.3**	**82.6**	**71.6**	**80.6**	**80.2**

Table 3
Comparing with SOTA methods on MPII validation set: 'Mean' is the average PCKh@0.5 score for all keypoints in all test images, and 'Mean0.1' is the average PCKh@0.1 score.

Method	Backbone	Input size	Hea	Sho	Elb	Wri	Hip	Kne	Ank	Mean	Mean0.1
Regression or Heatmap based											
SimpleBaseline [38]	ResNet-152	256x256	96.9	95.4	89.4	84.0	88.0	84.6	82.1	89.0	-
OKDHP-bran2 [18]	2-Stack HG	256x256	96.7	95.4	89.9	84.1	89.0	84.7	81.1	89.2	-
PRTR [15]	HRNet-W32	256x256	97.3	96.0	90.6	84.5	89.7	85.5	79.0	89.5	-
Simcc-based											
HRNet-W32 [32]	HRNet-W32	64x64	93.2	89.3	76.8	67.1	79.6	70.2	64.6	78.2	17.7
+Ours	HRNet-W32	64x64	**93.2**	89.2	**78.0**	**67.3**	**79.6**	**70.7**	**64.9**	**78.4**	**18.1**
HRNet-W32 [32]	HRNet-W32	256x256	96.8	95.8	89.9	85.0	88.9	85.1	80.9	89.4	36.6
+Ours	HRNet-W32	256x256	**97.0**	**96.1**	**89.9**	**85.1**	**89.3**	**85.6**	**81.3**	**89.6**	**37.0**



(a) 256×192 input size (b) 128×128 input size

(c) 256×192 input size (d) 128×128 input size

Fig. 3. Ablation study of coefficients for BL and SL under varying input resolutions. The first row represents the experimental results for the coefficient λ of BL at different values, while the second row represents the coefficient γ of SL. Dashed lines indicate the model's AP when BL and SL are not used. The experiments utilized HRNet-W32 as the backbone, and AP results are based on the COCO validation dataset.

Table 4
Ablation studies were performed on the Bilinear module, Bone loss, and Sum loss respectively. The baseline model is SimpleBaseline with ResNet50 as backbone.

Bilinear	Bone loss	Sum loss	AP	AP^M	AP^L
			73.8	71.0	78.6
✓			74.1(+0.3)	71.3	79.0
	✓		73.9(+0.1)	70.8	78.8
		✓	73.9(+0.1)	70.9	78.7
✓	✓		74.4(+0.6)	71.2	79.3
✓		✓	74.4(+0.6)	71.3	79.2

Table 5
Based on the comparative study of two loss functions Bone loss (BL) and Sum loss (SL) under different input sizes of HRNet-W32.

Method	Input size	AP	AP^M	AP^L
HRNet-W32		72.2	69.9	75.7
HRNet-W32+**BL**	128 × 128	72.5(+0.3)	70.1	76.4
HRNet-W32+**SL**		72.6(+0.4)	70.4	76.2
HRNet-W32		76.4	73.6	80.7
HRNet-W32+**BL**	256 × 192	76.9(+0.5)	74.2	81.1
HRNet-W32+**SL**		76.8(+0.4)	74.1	81.4

method. As shown in Table 5. The model has been validated in COCO val 2017 dataset. During training, the two added regularization losses allow the model to converge quickly. In the middle and late stages of model training, with the improvement of model fitting, the proportion of Sum loss in the total loss function is smaller than that of Bone loss in the same training period. Sum loss, however, had an unexpectedly positive outcome. When the input size is 256×192, the result of Bone loss addition outperforms HRNet-W32 0.5AP, and the result of Sum loss addition alone outperforms HRNet-W32 0.4AP. When the input size is 128×128, the Bone loss increases the accuracy by 0.3AP and the Sum loss increases by 0.4AP.

4.2.6. *The study of the loss weights*

As mentioned in Section 3.5, in the design of the overall loss function, weight coefficients are used to control the proportion of regularization terms within the overall loss function. The regularization terms play a supplementary role in keypoint localization, and if the weights are too large, they may interfere with the model's learning process. Therefore, during the model training process, we set these coefficients reasonably by observing the values of the original loss function and the two regularization terms. The Bone loss coefficient $\lambda \in \{1/5, 1/20, 1/35\}$ were tested on the basis of HRNet-W32. Fig. 3(a) shows that the performance of the model tends to increase and then decrease as $1/\lambda$ increases. For HRNet-W32, with a 256×192 input size, it is recommended to set $\lambda = 1/20$, which leads to an improvement of +0.5AP. The input size of 128×128, it brings an improvement of +0.3AP. For the coefficient γ of Sum loss, we tried $\gamma \in \{1, 1/2, 1/3\}$. By observing the experimental results, we recommend setting it to 1/2.

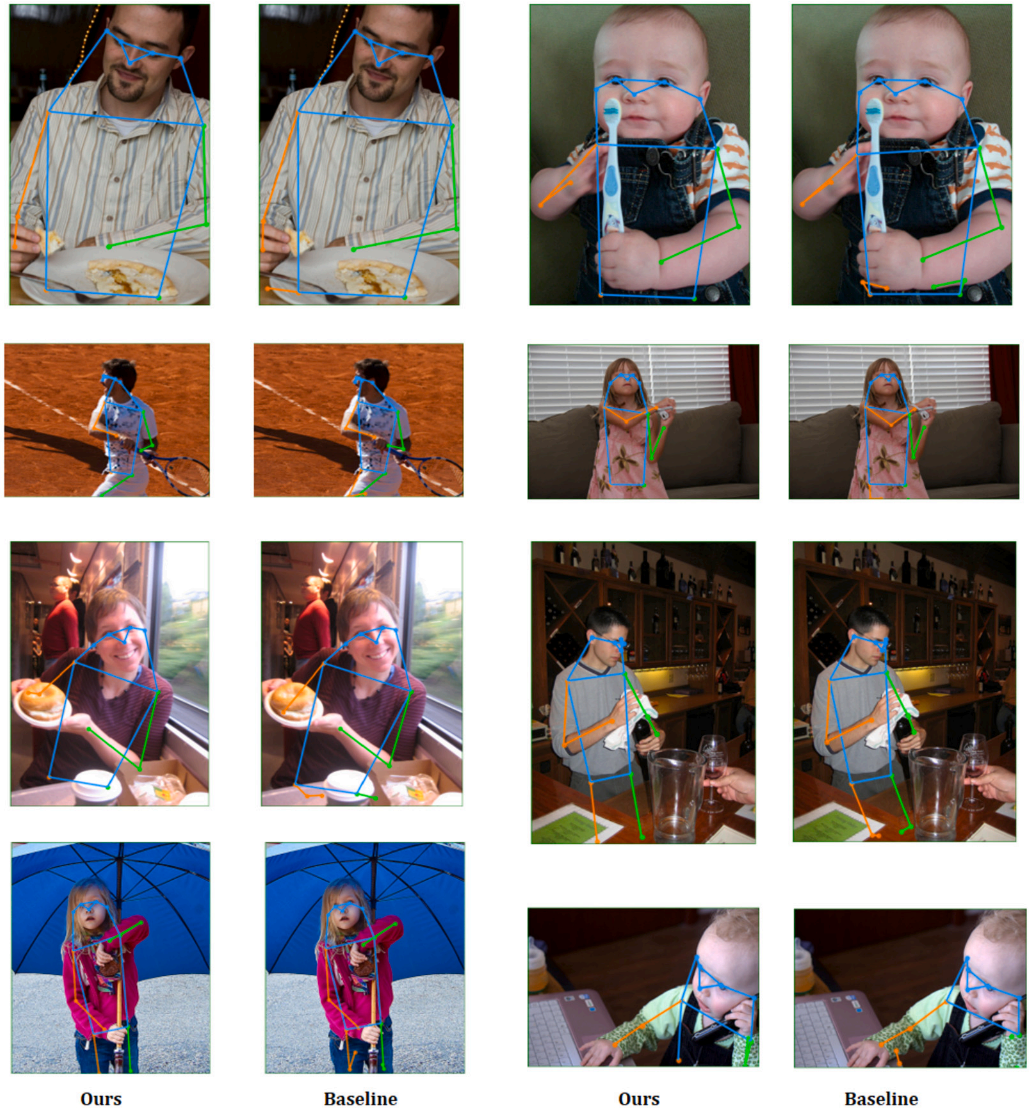


Fig. 4. Visualization of some predicted results when only part of the human body appears.

4.3. Prediction visualization

In Fig. 4, we show the non-full-body pictures in some half-body and occlusion cases. Our method is compared with the results predicted by some baseline methods, the left picture is the results using our method, and the right picture is the results predicted by the baseline model. The experimental results show that our method is able to help the model to limit the redundant predictions and make more reasonable estimates.

5. Conclusion

In this paper, we primarily investigate the resolution recovery module and loss function of existing 2D human pose estimation. In the resolution recovery part, an upsampling module with convolution plus bilinear interpolation is proposed to more accurately preserve features, which lays the groundwork for subsequent loss learning and coordinate decoding. For the human body constraint information, we design two regularization losses, namely Bone loss and Sum loss based on SimCC coordinate representation. These two regularization terms increase the error of the human keypoint constraint information in the loss function, so that the incorrectly predicted samples are punished more, thus reducing the generalization error. For this reason, our method is used on state-of-the-art models, and its performance is improved on COCO and MPII datasets.

CRediT authorship contribution statement

Yangqi Liu: Conceptualization, Methodology, Software. **Guodong Wang:** Conceptualization. **Hao Dong:** Software, Supervision. **Chenglizhao Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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